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Forward error correction (FEC) for dummy bearers in digital enhanced cordless telecommunications (DECT) ultra low energy (ULE) networks

Abstract

This disclosure provides methods, devices, and systems for wireless communications. The present implementations more specifically relate to packet designs that support forward error correction (FEC) coding for digital enhanced cordless telecommunications (DECT) ultra low energy (ULE) dummy bearers. In some aspects, an "enhanced" dummy bearer may include an S-field, an A-field, and a B-field, followed by one or more parity bits. The S-field carries a packet preamble and synchronization information which marks the start of a DECT packet transmission. The A-field carries control information associated with a DECT base station or network. The B-field carries control information associated with a ULE mode of operation by the base station. The parity bits may be combined with one or more fields (or subfields) of the dummy bearer to produce FEC codewords that can be decoded according to an FEC coding scheme.

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Background/Summary

TECHNICAL FIELD
(1) The present implementations relate generally to wireless communication, and specifically to forward error correction (FEC) for dummy bearers in digital enhanced cordless telecommunications (DECT) ultra low energy (ULE) networks.
BACKGROUND OF RELATED ART
(2) Digital enhanced cordless telecommunications (DECT) ultra low energy (ULE) is a wireless communication standard that can be used to implement home automation, security, and climate control in residential and enterprise environments. The basic building block of a ULE system is a home automation network (HAN) that includes a fixed part (FP) and at least one portable part (PP). A DECT ULE PP can be any portable device used in home automation, security, or climate control applications. Example suitable portable devices include, among other examples, sensors (such as smoke detectors or motion detectors), thermostats, and electricity control elements. The DECT ULE FP is a base station or access point that bridges a connection between the portable devices and a local network (or the Internet). The base station communicates with the portable devices over a DECT air interface. Each over-the-air (OTA) connection between the base station and a respective portable device is referred to as a wireless communication “link.”
(3) A base station periodically transmits or broadcasts “dummy bearers” (also referred to as “beacons”) to enable any portable devices within wireless communication range of the base station to establish or maintain a communication link with the base station or network. For example, a base station may transmit a dummy bearer every 10 ms. Dummy bearers are network packets that carry control information which can be used by a portable device to identify the base station and synchronize to it. Dummy bearers also may carry paging information that can be used by a portable device to receive incoming messages or connections from the base station.
(4) Imperfect channel conditions on the DECT air interface may result in data loss during the transmission of dummy bearers. In existing DECT ULE networks, a portable device that receives incomplete (or incorrect) dummy bearer information may be unable to recover the dummy bearer using the received information. Accordingly, the portable device discards the received dummy bearer information and waits for the base station to broadcast another dummy bearer, resulting in a waste of the portable device's resources (including memory, power, and processing resources). Thus, there is a need to improve the ability of a portable device to receive dummy bearers from a base station under imperfect channel conditions.
SUMMARY
(5) This Summary is provided to introduce in a simplified form a selection of concepts that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to limit the scope of the claimed subject matter.
(6) One innovative aspect of the subject matter of this disclosure can be implemented in a method of wireless communication performed by a wireless communication device. The method includes receiving, over a wireless communication channel, at least a portion of a dummy bearer that includes a B-field followed by one or more first parity bits, the B-field carrying control information associated with an ultra low energy (ULE) mode of operation by a digital enhanced cordless telecommunications (DECT) base station; constructing a first codeword that includes at least part

of the B-field and the one or more first parity bits; and recovering the control information based on the first codeword and a forward error correction (FEC) code associated with the first codeword.
(7) Another innovative aspect of the subject matter of this disclosure can be implemented in a wireless communication device, including a processing system and a memory. The memory stores instructions that, when executed by the processing system, cause the wireless communication device to receive, over a wireless communication channel, at least a portion of a dummy bearer that includes a B-field followed by one or more first parity bits, the B-field carrying control information associated with a ULE mode of operation by a DECT base station; construct a first codeword that includes at least part of the B-field and the one or more first parity bits; and recover the control information based on the first codeword and a FEC code associated with the first codeword.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present implementations are illustrated by way of example and are not intended to be limited by the figures of the accompanying drawings.
(2) FIG. **1** shows an example wireless communication network.
(3) FIG. **2** shows an example dummy bearer usable for wireless communications between a base station and a portable device.
(4) FIG. **3** shows a block diagram of an example communication system that supports forward error correction (FEC) coding.
(5) FIG. **4** shows an enhanced dummy bearer usable for wireless communications between a base station and a portable device, according to some implementations.
(6) FIG. **5** shows another enhanced dummy bearer usable for wireless communications between a base station and a portable device, according to some implementations.
(7) FIGS. **6**A and **6**B show timing diagrams depicting example communications between a base station and a number of portable devices, according to some implementations.
(8) FIG. **7** shows another enhanced dummy bearer usable for wireless communications between a base station and a portable device, according to some implementations.
(9) FIG. **8**A shows another enhanced dummy bearer usable for wireless communications between a base station and a portable device, according to some implementations.
(10) FIG. **8**B shows another enhanced dummy bearer usable for wireless communications between a base station and a portable device, according to some implementations.
(11) FIG. **8**C shows another enhanced dummy bearer usable for wireless communications between a base station and a portable device, according to some implementations.
(12) FIG. **8**D shows another enhanced dummy bearer usable for wireless communications between a base station and a portable device, according to some implementations.
(13) FIG. **9** shows a block diagram of an example wireless communication device, according to some implementations.
(14) FIG. **10** shows an illustrative flowchart depicting an example operation for wireless communications, according to some implementations.
DETAILED DESCRIPTION
(15) In the following description, numerous specific details are set forth such as examples of specific components, circuits, and processes to provide a thorough understanding of the present disclosure. The term “coupled” as used herein means connected directly to or connected through one or more intervening components or circuits. The terms “electronic system” and “electronic device” may be used interchangeably to refer to any system capable of electronically processing information. Also, in the following description and for purposes of explanation, specific nomenclature is set forth to provide a thorough understanding of the aspects of the disclosure.

However, it will be apparent to one skilled in the art that these specific details may not be required to practice the example embodiments. In other instances, well-known circuits and devices are shown in block diagram form to avoid obscuring the present disclosure. Some portions of the detailed descriptions which follow are presented in terms of procedures, logic blocks, processing and other symbolic representations of operations on data bits within a computer memory.

(16) These descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. In the present disclosure, a procedure, logic block, process, or the like, is conceived to be a self-consistent sequence of steps or instructions leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, although not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated in a computer system. It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities.

(17) Unless specifically stated otherwise as apparent from the following discussions, it is appreciated that throughout the present application, discussions utilizing the terms such as “accessing,” “receiving,” “sending,” “using,” “selecting,” “determining,” “normalizing,” “multiplying,” “averaging,” “monitoring,” “comparing,” “applying,” “updating,” “measuring,” “deriving” or the like, refer to the actions and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

(18) In the figures, a single block may be described as performing a function or functions; however, in actual practice, the function or functions performed by that block may be performed in a single component or across multiple components, and/or may be performed using hardware, using software, or using a combination of hardware and software. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described below generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure. Also, the example input devices may include components other than those shown, including well-known components such as a processor, memory and the like.

(19) The techniques described herein may be implemented in hardware, software, firmware, or any combination thereof, unless specifically described as being implemented in a specific manner. Any features described as modules or components may also be implemented together in an integrated logic device or separately as discrete but interoperable logic devices. If implemented in software, the techniques may be realized at least in part by a non-transitory processor-readable storage medium including instructions that, when executed, performs one or more of the methods described above. The non-transitory processor-readable data storage medium may form part of a computer program product, which may include packaging materials.

(20) The non-transitory processor-readable storage medium may comprise random access memory (RAM) such as synchronous dynamic random-access memory (SDRAM), read only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, other known storage media, and the like. The techniques additionally, or alternatively, may be realized at least in part by a processor-readable communication medium that carries or communicates code in the form of instructions or data structures and that can be accessed, read, and/or executed by a computer or other processor.

(21) The various illustrative logical blocks, modules, circuits and instructions described in connection with the embodiments disclosed herein may be executed by one or more processors (or a processing system). The term "processor," as used herein may refer to any general-purpose processor, special-purpose processor, conventional processor, controller, microcontroller, and/or state machine capable of executing scripts or instructions of one or more software programs stored in memory.
(22) As described above, base stations in digital enhanced cordless telecommunications (DECT) ultra low energy (ULE) networks periodically transmit or broadcast "dummy bearers" (also referred to as "beacons") to enable any portable devices within wireless communication range to establish or maintain a communication link with the base station or network. However, imperfect channel conditions on the DECT air interface may result in data loss during the transmission of dummy bearers. Aspects of the present disclosure recognize that forward error correction (FEC) codes (also referred to as "error-correcting codes") can compensate for the intrinsic unreliability of information transfer in wireless communication systems by introducing redundancy into the data stream. Example suitable FEC codes include Reed-Solomon codes, turbo codes, and low-density parity-check (LDPC) codes, among other examples. Existing versions of the DECT ULE standard do not support FEC coding for dummy bearers. Thus, new packet designs are needed to enable FEC coding for the transmission and reception of dummy bearers in a DECT ULE network.
(23) Various aspects relate generally to DECT ULE networks, and more particularly, to packet designs that support FEC coding for DECT ULE dummy bearers. In some aspects, an "enhanced" dummy bearer may include an S-field, an A-field, and a B-field, followed by one or more parity bits. The S-field carries a packet preamble and synchronization information which marks the start of a DECT packet transmission. The A-field carries control information associated with a DECT base station or network (such as an identity of the base station, one or more capabilities of the base station, whether the base station currently supports new subscriptions, and page messages indicating incoming connections). The B-field carries control information associated with a ULE mode of operation by the base station (such as channel selection information and page messages indicating incoming connections). The parity bits may be combined with one or more fields (or subfields) of the dummy bearer to produce FEC codewords that can be decoded according to an FEC coding scheme.
(24) In some aspects, a portable device may recover or reproduce portions of the dummy bearer that were lost during transmission (such as bits that were erased or altered) based on the decoded FEC codewords. In some implementations, the parity bits may be associated with the B-field of the dummy bearer. In such implementations, the parity bits may be used to recover the control information carried in the B-field. In some other implementations, the parity bits may be associated with the A-field of the dummy bearer. In such implementations, the parity bits may be used to recover the control information carried in the A-field. Still further, in some implementations, the parity bits may include one or more first parity bits associated with the B-field followed by one or more second parity bits associated with the A-field. In such implementations, the first parity bits may be used to recover the control information carried in the B-field and the second parity bits may be used, independent of the first parity bits, to recover the control information carried in the A-field.
(25) Particular implementations of the subject matter described in this disclosure can be implemented to realize one or more of the following potential advantages. By adding FEC coding to DECT ULE dummy bearers, aspects of the present disclosure can improve the robustness of the dummy bearers to imperfect channel conditions on the DECT air interface. For example, adding parity bits to a dummy bearer allows a portable device to recover control information that would otherwise be lost during transmission. As a result, the portable device may conserve memory, power, and processing resources that would otherwise be used to receive another transmission of the dummy bearer. By adding the parity bits to the end of the dummy bearer (after the A-field and

the B-field), the packet formats of the present disclosure may be backwards compatible with legacy devices. As used herein, the term “legacy” device refers to any portable device (or base station) that conforms to existing versions of the DECT ULE standard and does not support FEC coding of dummy bearers. Legacy devices may terminate reception of a dummy bearer once the A-field or the B-field has been received, and thus not receive the parity bits.
(26) FIG. **1** shows an example wireless communication network **100**. In some aspects, the wireless communication network **100** may be one example of a home automation network (HAN) conforming to the digital enhanced cordless telecommunications (DECT) ultra low energy (ULE) wireless communication standard. The network **100** includes a gateway **110** and a portable device **120**. Although a single portable device **120** is shown in the example of FIG. **1** (for simplicity), the network **100** may include any number of portable devices.
(27) The gateway **110** includes a host processor **112** and a base station **114**. The base station **114** (also referred to as a “root node”) represents a fixed part (FP) of the network **100**. More specifically, the base station **114** enables the host processor **112** to communicate with portable devices in the network **100** (such as the portable device **120**) over a DECT air interface. For example, the base station **114** may transmit downlink messages, via the DECT air interface, to the portable device **120** on behalf of the host processor **112**. The base station **114** also may receive uplink messages, via the DECT air interface, from the portable device **120** and provide the received messages to the host processor **112**.
(28) The portable device **120** (also referred to as a “leaf”) represents a portable part (PP) of the network **100**. In some implementations, the portable device **120** may support a ULE mode of operation defined by the DECT ULE standard. Portable devices that support the ULE mode are referred to herein as “ULE devices.” A ULE device can be any portable device that is used in home automation, security, or climate control applications. Example suitable portable devices may include sensors (such as smoke detectors or motion sensors), thermostats, and electricity control elements, among other examples. In some other implementations, the portable device **120** may not support the ULE mode of operation. Portable devices that do not support the ULE mode may be referred to herein as “DECT devices.”
(29) The base station **114** periodically transmits or broadcasts dummy bearers **102** (also referred to as “beacons”) over the DECT air interface. For example, existing versions of the DECT ULE standard require each base station to broadcast a dummy bearer every 10 ms. The dummy bearer **102** is a network packet that carries control information which can be used by any portable devices within wireless communication range of the base station **114** (such as the portable device **120**) to identify the base station **114** and synchronize to it. The dummy bearer **102** also may carry paging information which can be used by portable devices that are synchronized with the base station **114** to receive incoming messages or connections from the base station **114**.
(30) FIG. **2** shows an example dummy bearer **200** usable for wireless communications between a base station and a portable device. More specifically, the dummy bearer **200** conforms to a packet format defined by existing versions of the DECT ULE standard (also referred to as a “legacy” dummy bearer packet format).
(31) The dummy bearer **200** includes an S-field **210**, an A-field **220**, a B-field **230**, an X-field **240**, and a Z-field **250**. The S-field **210** carries a packet preamble and synchronization information which marks the start of a DECT packet transmission. All portable devices, including DECT devices and ULE devices, receive the S-field **210** of at least one dummy bearer broadcast by a base station (such as during an initial subscription or service call procedure, among other examples) to prepare or configure their receivers to receive the remainder of the dummy bearer **200** (such as by adjusting the receiver frequency or recovering signal timing information). Unlike DECT devices, which receive the S-field **210** of each dummy bearer **200** transmitted by a base station, ULE devices may ignore the S-field **210** of subsequent dummy bearers **200** received during a “normal” operation (such as for incoming connection notifications).

(32) The A-field **220** carries control information associated with a base station or network (such as an identity of the base station, one or more capabilities of the base station, whether the base station currently supports new subscriptions, and page messages indicating incoming connections). All portable devices, including DECT devices and ULE devices, also receive the A-field **220** of at least one dummy bearer broadcast by a base station (such as during an initial subscription or service call procedure, among other examples) to acquire current network information (such as available timeslots for transmission or information about incoming connections). Unlike DECT devices, which receive the A-field **220** of each dummy bearer **200** transmitted by a base station, ULE devices may ignore the A-field **220** of subsequent dummy bearers **200** received during a “normal” operation (such as for incoming connection notifications).
(33) The B-field **230** carries control information associated with a ULE mode of operation by the base station. Thus, only ULE devices may receive or interpret the B-field **230** of the dummy bearer **200**, whereas DECT devices may terminate their reception of the dummy bearer **200** after receiving the A-field **220**. According to existing versions of the DECT ULE standard, the B-field **230** is subdivided into four subfields. The first subfield (Subfield 0) signals the start of the B-field **230** and may include information that identifies the base station. The remaining subfields (Subfield 1, Subfield 2, and Subfield 3) carry the control information relevant for ULE operation (such as channel selection information and page messages indicating incoming connections).
(34) ULE devices may use the information in Subfield 0 to prepare or configure their receivers to receive the remainder of the dummy bearer **200** (such as by adjusting the receiver frequency or recovering signal timing information) and to verify whether the dummy bearer **200** is associated with a desired base station. Thus, during normal operation, ULE devices may begin receiving the dummy bearer **200** starting from the B-field **230** (such as by detecting Subfield 0). ULE devices may further use the control information carried in Subfields 1-3 to set up connections for ULE data transfers and to check for incoming connections.
(35) The X-field **240** carries cyclic redundancy check (CRC) information generated over parts of the B-field **230**, and the Z-field **250** is a duplication of the X-field **240**. Portable devices (such as DECT devices and ULE devices) may use the CRC information in the X-field **240** to determine whether the B-field **230** contains any errors and may use the Z-field **250** to detect interference at the end of the dummy bearer **200**. However, many DECT devices ignore the B-field **230** entirely and may thus also ignore the X-field **240** and the Z-field **250**. Moreover, existing versions of the DECT ULE standard do not support mechanisms for notifying the base station of any detected interference. As a result, some ULE devices also ignore the Z-field **250**.
(36) Imperfect channel conditions on the DECT air interface may result in data loss during the transmission of dummy bearers. For example, one or more bits of the dummy bearer **200** may be erased or altered when the dummy bearer **200** reaches a portable device (such as the portable device **120** of FIG. **1**). Aspects of the present disclosure recognize that forward error correction (FEC) codes (also referred to as “error-correcting codes”) can compensate for the intrinsic unreliability of information transfer in wireless communication systems by introducing redundancy into the data stream. In some aspects, a base station may encode at least a portion of a dummy bearer according to an FEC coding scheme. A portable device may correct errors in the received data by decoding the dummy bearer according to the FEC coding scheme.
(37) FIG. **3** shows a block diagram of an example communication system **300** that supports FEC coding. The communication system **300** includes a FEC encoder **310**, a channel **320**, and a FEC decoder **330**. The FEC encoder **310** may be included in, or correspond to, a transmitting device. In some implementations, the transmitting device may be one example of the base station **114** of FIG. **1**. The decoder **320** may be included in, or correspond to, a receiving device. In some implementations, the receiving device may be one example of the portable device **120** of FIG. **1**.
(38) The channel **320** may include any communication link between the encoder **310** and the decoder **330**. In some implementations, the channel **320** may be a wireless communication medium

(such as a DECT air interface). Imperfections in the channel **320** may introduce channel distortion (such as linear distortion, multi-path effects, Additive White Gaussian Noise (AWGN), and the like). To compensate for channel imperfections, the encoder **310** may encode (TX) data **301** to be transmitted over the channel **320** so that error correction may be performed by the decoder **330** to recover the (RX) data **302**.
(39) In some implementations, the FEC encoder **310** may use one or more FEC codes to encode the TX data **301** into one or more TX codewords (CWs) **312**. Example suitable FEC codes include Reed-Solomon codes, turbo codes, and low-density parity-check (LDPC) codes, among other examples. The FEC decoder **330** may use the same FEC codes to decode one or more RX codewords (CWs) **322**, received via the channel **320**, and recover the RX data **302**. If the channel **320** introduces errors (such as erased or altered bits) into the TX codewords **312**, the decoder **330** may detect and correct such errors in the RX codewords **322**, according to the FEC code, so that the RX data **302** matches the TX data **301**.
(40) The FEC encoding operation adds one or more parity bits to the TX data **301** (such that each TX codeword **312** includes the TX data **301** plus the one or more parity bits). In some implementations, the TX data **301** may include one or more portions of a dummy bearer (such as the dummy bearer **200** of FIG. **2**). However, existing versions of the DECT ULE standard do not support FEC coding for dummy bearers. Thus, new packet designs are needed to enable FEC coding for the transmission and reception of dummy bearers in a DECT ULE network.
(41) FIG. **4** shows an enhanced dummy bearer **400** usable for wireless communications between a base station and a portable device, according to some implementations. In some implementations, the enhanced dummy bearer **400** may be one example of the dummy bearer **102** of FIG. **1**.
(42) The enhanced dummy bearer **400** includes an S-field **410**, an A-field **420**, a B-field **430**, an X-field **440**, a Z-field **450**, and parity information **432**. In some implementations, the S-field **410**, A-field **420**, and B-field **430** may be examples of the S-field **210**, A-field **220**, and B-field **230**, respectively, of FIG. **2**. For example, the S-field **410** may carry a packet preamble and synchronization information which marks the start of a DECT packet transmission; the A-field **420** may carry control information associated with a base station or network; and the B-field **430** may carry control information associated with a ULE mode of operation by the base station. In some implementations, the X-field **440** may carry CRC information generated over parts of the B-field **430**, and the Z-field **450** may be a duplication of the X-field **440**.
(43) In some aspects, the parity information **432** may be used to correct errors in the enhanced dummy bearer **400** according to an FEC coding scheme. For example, the parity information **432** may be combined with one or more portions of the enhanced dummy bearer **400** to produce FEC codewords that can be decoded, using the FEC coding scheme, to recover bits that were erased or altered during transmission over a DECT air interface (such as described with reference to FIG. **3**). Example suitable FEC codes include Reed-Solomon codes, turbo codes, and LDPC codes, among other examples. In some implementations, the parity information **432** may be used to recover the control information carried in the B-field **430**. In some other implementations, the parity information **432** may be used to recover the control information carried in the A-field **420**. Still further, in some implementations, the parity information **432** may be used to recover any of the control information carried in the A-field **420** or the B-field **430**.
(44) Aspects of the present disclosure recognize that some DECT networks may include legacy portable devices (including DECT devices and legacy ULE devices) that do not support FEC decoding for dummy bearers. Thus, in some implementations, the parity information **432** may be added after the B-field **430** so that the enhanced dummy bearer **400** may be backwards compatible with legacy portable devices. For example, the parity information **432** may be included in an extended B-field payload that conforms with existing DECT packet formats (such as the DECT long slot format). With reference for example to FIG. **2**, the first three fields **410-430** of the enhanced dummy bearer **400** are the same as the first three fields **210-230** of the dummy bearer

200. Thus, DECT devices may terminate reception of the enhanced dummy bearer **400** after receiving the A-field **420**, and legacy ULE devices **430** may terminate reception of the enhanced dummy bearer **400** after receiving the B-field **430**.

(45) FIG. **5** shows another enhanced dummy bearer **500** usable for wireless communications between a base station and a portable device, according to some implementations. In some implementations, the enhanced dummy bearer **500** may be one example of the enhanced dummy bearer **400** of FIG. **4**.

(46) The enhanced dummy bearer **500** includes an S-field **510**, an A-field **520**, a B-field **530**, a first X-field **540**, a first Z-field **550**, a second X-field **560**, a second Z-field **570**, one or more parity bits **532** associated with the B-field **530** (also referred to as “B” parity bits), and one or more parity bits **522** associated with the A-field **520** (also referred to as “A” parity bits). With reference to FIG. **4**, the S-field **510**, A-field **520**, B-field **530**, second X-field **560**, and second Z-field **570** may be examples of the S-field **410**, A-field **420**, B-field **430**, X-field **440**, and Z-field **450**, respectively, of the enhanced dummy bearer **400**. For example, the S-field **510** may carry a packet preamble and synchronization information which marks the start of a DECT packet transmission; the A-field **520** may carry control information associated with a base station or network; the B-field **530** may carry control information associated with a ULE mode of operation by the base station; the X-field **560** may carry CRC information generated over parts of the B-field **530**; and the Z-field **570** may be a duplication of the X-field **560**.

(47) In some implementations, the first X-field **540** and the first Z-field **550** may provide backwards compatibility with legacy devices. With reference for example to FIG. **2**, the first five fields **510-550** of the enhanced dummy bearer **500** match the first five fields **210-250** of the dummy bearer **200**, which conforms with existing DECT ULE standards. In such implementations, the X-field **540** may carry CRC information generated over parts of the B-field **530** and the Z-field **550** may be a duplicate of the X-field **540**. However, as described with reference to FIG. **2**, many legacy portable devices ignore the X-field **240** or the Z-field **250**. Thus, in some other implementations, at least one of the first X-field **540** or the first Z-field **550** may be repurposed to carry other information (such as signaling information for the enhanced dummy bearer **500**).

(48) The B parity bits **532** and the A parity bits **522** may be examples of the parity information **432** of FIG. **4**. In some implementations, the B parity bits **532** may be combined with one or more portions of the B-field **530** to produce one or more FEC codewords **502** that can be decoded according to an FEC decoding scheme, and the A parity bits **522** may be combined with one or more portions of the A-field **520** to produce one or more FEC codewords **504** that can be decoded according to the same (or a different) FEC coding scheme. Example suitable FEC codes include Reed-Solomon codes, turbo codes, and LDPC codes, among other examples. With reference to FIG. **3**, each of the FEC codewords **502** and **504** may be one example of the TX codewords **312** (or RX codewords **322**). More specifically, the A parity bits **522** may be used to recover control information carried in the A-field **520** and the B parity bits **532** may be used to recover control information carried in the B-field **530**.

(49) In some implementations, the B parity bits **532** also may be used to recover information carried in the first X-field **540** or the first Z-field **550**. For example, the FEC codewords formed by the B parity bits **532** may include the first X-field **540** or the first Z-field **550** (in addition to one or more portions of the B-field **530**). In some other implementations, the B parity bits **532** may not be used to recover any information carried in the first X-field **540** or the first Z-field **550** (such as to improve FEC decoding for the B-field **530**). With reference for example to FIG. **2**, the first subfield (Subfield 0) of the B-field **530** remains the same or constant across all dummy bearers broadcast by the same base station. Thus, a portable device already knows the information carried in Subfield 0 when receiving the enhanced dummy bearer **500** during normal operation. In some implementations, Subfield 0 of the B-field **530** may not be encoded (or decoded) using FEC coding. In other words, the B parity bits **532** may not be used to recover any of the information in

Subfield 0.
(50) As shown in FIG. **5**, the B parity bits **532** are transmitted (and received) immediately following the Z-field **550**, and the A parity bits **522** are transmitted (and received) immediately following the B parity bits **532**. The example arrangement of parity bits **532** and **522** may allow ULE devices to minimize their time of reception for the enhanced dummy bearer **500** during normal operation. As described with reference to FIG. **2**, a ULE device may begin receiving the enhanced dummy bearer **500** starting at the B-field **530**, during normal operation, thereby ignoring the A-field **520**. By inserting the B parity bits **532** before the A parity bits **522** in the enhanced dummy bearer **500**, aspects of the present disclosure may allow a ULE device to correct errors in the B-field **530** (if necessary) using the B parity bits **532** and terminate reception of the enhanced dummy bearer **500** prior to receiving the A parity bits **522**.
(51) FIG. **6**A shows a timing diagram **600** depicting an example communication between a base station **602** and a number of portable devices **604-608**, according to some implementations. In some implementations, the base station **602** may be one example of the base station **114** of FIG. **1**, and each of the portable devices **604-608** may be one example of the portable device **120**.
(52) The base station **602** broadcasts a dummy bearer **601**, between times t.sub.0 and t.sub.4, which is received by a DECT device **604**, a legacy ULE device **606**, and a ULE device **608**. In some implementations, the dummy bearer **601** may be one example of the enhanced dummy bearer **500** of FIG. **5**. More specifically, the dummy bearer **601** includes an S-field, an A-field, a B-field, a first X-field, a first Z-field, one or more B parity bits following the first Z-field, one or more A parity bits following the B parity bits, a second X-field following the A parity bits, and a second Z-field following the second X-field.
(53) The DECT device **604** does not support a ULE mode of operation by the base station **602**. Thus, as shown in FIG. **6**A, the DECT device **604** receives only the S-field and the A-field of the dummy bearer **601**. In other words, the DECT device **604** may terminate its reception of the dummy bearer **601**, at time t.sub.1, prior to receiving the B-field. To conserve power, a portable device (such as any of the portable devices **604**, **606**, or **608**) may turn off or otherwise deactivate one or more of its wireless radios upon completing or terminating reception of the dummy bearer **601**.
(54) The legacy ULE device **606** supports the ULE mode of operation but does not support FEC decoding for dummy bearers. In the example of FIG. **6**A, the legacy ULE device **606** receives the dummy bearer **601** during an initial subscription or service call procedure. Thus, the legacy ULE device **606** receives the S-field, A-field, and B-field of the dummy bearer **601**. In some implementations, the legacy ULE device **606** may terminate its reception of the dummy bearer **601**, at time t.sub.2, prior to receiving the first X-field (as shown in FIG. **6**A). In some other implementations, the legacy ULE device **606** may terminate its reception of the dummy bearer **601**, at time t.sub.1, prior to receiving the B-field (such as during an initial subscription). Still further, in some implementations, the legacy ULE device **606** may receive the first X-field and the first Z-field before terminating its reception of the dummy bearer **601**.
(55) The ULE device **608** supports the ULE mode of operation and also supports FEC decoding for dummy bearers. In the example of FIG. **6**A, the ULE device **608** receives the dummy bearer **601** during an initial subscription or service call procedure. Thus, the ULE device **608** receives the S-field, A-field, and B-field of the dummy bearer **601**. In some implementations, the ULE device **608** may further receive the first X-field, the first Z-field, the B-parity bits, and the A-parity bits of the dummy bearer **601** (as shown in FIG. **6**A). In some other implementations, the ULE device **608** may terminate its reception of the dummy bearer **601** prior to receiving the B parity bits or the A parity bits (such as when the CRC information carried in the A-field, the B-field, or the X-field indicates that the received data does not contain any errors).
(56) In some implementations, the ULE device **608** may terminate its reception of the dummy bearer **601**, at time t.sub.3, prior to receiving the second X-field (as shown in FIG. **6**A). In some

other implementations, the ULE device **608** may receive the dummy bearer **601** in its entirety (including the second X-field and the second Z-field). In some other implementations, the ULE device **608** may terminate its reception of the dummy bearer **601**, at time t.sub.2, prior to receiving the first X-field (such as when the CRC information carried in the B-field indicates that the received data does not contain any errors). Still further, in some implementations, the ULE device **608** may terminate its reception of the dummy bearer **601**, at time t.sub.1, prior to receiving the B-field (such as during an initial subscription).

(57) FIG. **6**B shows another timing diagram **610** depicting an example communication between the base station **602** and the portable devices **604-608** of FIG. **6**A, according to some implementations. In the example of FIG. **6**B, the legacy ULE device **606** and the ULE device **608** are configured to operate in a normal mode after the completing the initial subscription or service call procedure described with reference to FIG. **6**A.

(58) The base station **602** retransmits the dummy bearer **601**, between times t.sub.5 and t.sub.9, which is received by the DECT device **604**, the legacy ULE device **606**, and the ULE device **608**. In some implementations, the dummy bearer **601** may be one example of the enhanced dummy bearer **500** of FIG. **5**. More specifically, the dummy bearer **601** includes an S-field, an A-field, a B-field, a first X-field, a first Z-field, one or more B parity bits following the first Z-field, one or more A parity bits following the B parity bits, a second X-field following the A parity bits, and a second Z-field following the second X-field.

(59) In the example of FIG. **6**B, the DECT device **604** receives only the S-field and the A-field of the dummy bearer **601** and the legacy ULE device **606** receives only the B-field of the dummy bearer **601**. In other words, the DECT device **604** may terminate its reception of the dummy bearer **601**, at time t.sub.6, prior to receiving the B-field, whereas the legacy ULE device **606** may not begin receiving the dummy bearer **601** until time t.sub.6. In some implementations, the legacy ULE device **606** may terminate its reception of the dummy bearer **601**, at time t.sub.7, prior to receiving the first X-field (as shown in FIG. **6**B). In some other implementations, the legacy ULE device **606** may receive the first X-field and the first Z-field before terminating its reception of the dummy bearer **601**.

(60) The ULE device **608** also receives the dummy bearer **601**, starting with the B-field, at time t.sub.6. In some implementations, the ULE device **608** may further receive the first X-field, the first Z-field, and the B parity bits of the dummy bearer **601** (as shown in FIG. **6**B). Because the ULE device **608** ignores the A-field of the dummy bearer **601**, the ULE device **608** also may ignore the A parity bits. Thus, the ULE device **608** may terminate its reception of the dummy bearer **601**, at time t.sub.8, prior to receiving the A parity bits (as shown in FIG. **6**B). In some other implementations, the ULE device **608** may terminate its reception of the dummy bearer **601**, at time t.sub.7, prior to receiving the X-field or the B parity bits (such as when the CRC information carried in the B-field, or the X-field, indicates that the received data does not contain any errors).

(61) Aspects of the present disclosure recognize that, depending on the FEC coding scheme used to encode the A-field or the B-field, the total number of parity bits added to an enhanced dummy bearer (such as the A parity bits plus the B parity bits) may be less than the additional overhead provided by the DECT long slot format (or other DECT packet format used to support the additional parity information). As a result, some enhanced dummy bearers may have one or more spare or unused bits leftover after the parity bits are added. In some implementations, at least some of the spare bits in an enhanced dummy bearer may be repurposed to carry other information (such as signaling information for the enhanced dummy bearer).

(62) FIG. **7** shows another enhanced dummy bearer **700** usable for wireless communications between a base station and a portable device, according to some implementations. In some implementations, the enhanced dummy bearer **700** may be one example of the enhanced dummy bearer **400** of FIG. **4**.

(63) The enhanced dummy bearer **700** includes an S-field **710**, an A-field **720**, a B-field **730**, a first

X-field **740**, a first Z-field **750**, a second X-field **760**, a second Z-field **770**, one or more “B” parity bits **732**, one or more “A” parity bits **722**, and one or more additional bits **780**. With reference to FIG. **4**, the S-field **710**, A-field **720**, B-field **730**, second X-field **760**, and second Z-field **770** may be examples of the S-field **410**, A-field **420**, B-field **430**, X-field **440**, and Z-field **450**, respectively, of the enhanced dummy bearer **400**. For example, the S-field **710** may carry a packet preamble and synchronization information which marks the start of a DECT packet transmission; the A-field **720** may carry control information associated with a base station or network; the B-field **730** may carry control information associated with a ULE mode of operation by the base station; the X-field **760** may carry CRC information generated over parts of the B-field **730**; and the Z-field **770** may be a duplication of the X-field **760**.

(64) In some implementations, the first X-field **740** and the first Z-field **750** may provide backwards compatibility with legacy devices. With reference for example to FIG. **2**, the first five fields **710-750** of the enhanced dummy bearer **700** match the first five fields **210-250** of the dummy bearer **200**, which conforms with existing DECT ULE standards. In such implementations, the X-field **740** may carry CRC information generated over parts of the B-field **730** and the Z-field **750** may be a duplicate of the X-field **740**. However, as described with reference to FIG. **2**, many legacy portable devices ignore the X-field **240** or the Z-field **250**. Thus, in some other implementations, at least one of the first X-field **740** or the first Z-field **750** may be repurposed to carry other information (such as signaling information for the enhanced dummy bearer **700**).

(65) The B parity bits **732** and the A parity bits **722** may be examples of the parity information **432** of FIG. **4**. In some implementations, the B parity bits **732** may be combined with one or more portions of the B-field **730** to produce one or more FEC codewords **702** that can be decoded according to an FEC decoding scheme, and the A parity bits **722** may be combined with one or more portions of the A-field **720** to produce one or more FEC codewords **704** that can be decoded according to the same (or a different) FEC coding scheme. With reference to FIG. **3**, each of the FEC codewords **702** and **704** may be one example of the TX codewords **312** (or RX codewords **322**). More specifically, the A parity bits **722** may be used to recover control information carried in the A-field **720** and the B parity bits **732** may be used to recover control information carried in the B-field **730**. In some implementations, the B parity bits **732** may not be used to recover information carried in at least one of the subfields (such as Subfield 0) of the B-field **730**.

(66) In some aspects, the additional bits **780** may carry signaling information associated with the enhanced dummy bearer **700**. As shown in FIG. **7**, the additional bits **780** are transmitted (and received) immediately following the Z-field **750**, the B parity bits **732** are transmitted (and received) immediately following the additional bits **780**, and the A parity bits **722** are transmitted (and received) immediately following the B parity bits **732**. In some implementations, the additional bits **780** may be merged or combined with at least one of the first X-field **740** or the first Z-field **750** to carry additional signaling information (such as by repurposing the first X-field **740** or the first Z-field **750** to carry signaling information rather than CRC information).

(67) In some implementations, the B parity bits **732** also may be used to recover information carried in the first X-field **740**, the first Z-field **750**, or the additional bits **780**. For example, the FEC codewords formed by the B parity bits **732** may include the first X-field **740**, the first Z-field **750**, or the additional bits **780** (in addition to one or more portions of the B-field **730**). In some other implementations, the B parity bits **732** may not be used to recover any of the information carried in the first X-field **740**, the first Z-field **750**, or the additional bits **780**.

(68) FIG. **8**A shows another enhanced dummy bearer **800** usable for wireless communications between a base station and a portable device, according to some implementations. In some implementations, the enhanced dummy bearer **800** may be one example of the enhanced dummy bearer **400** of FIG. **4**.

(69) The enhanced dummy bearer **800** includes an S-field **801**, an A-field **802**, one or more “A” parity bits **803**, a B-field **804**, one or more “B” parity bits **805**, and X- and Z-fields **806**. With

reference to FIG. **4**, the S-field **801**, A-field **802**, and B-field **804** may be examples of the S-field **410**, A-field **420**, and B-field **430**, respectively, of the enhanced dummy bearer **400**. The X- and Z-fields **806** may be examples of the X-field **440** and the Z-field **450** (shown as combined block, in FIG. **8**A, for simplicity). Thus, the S-field **801** may carry a packet preamble and synchronization information which marks the start of a DECT packet transmission; the A-field **802** may carry control information associated with a base station or network; the B-field **804** may carry control information associated with a ULE mode of operation by the base station; and the X- and Z-fields **806** may carry CRC information generated over parts of the B-field **804**.

(70) The A parity bits **803** and the B parity bits **805** may be examples of the parity information **432** of FIG. **4**. In some implementations, the B parity bits **805** may be combined with one or more portions of the B-field **804** to produce one or more FEC codewords **807** that can be decoded according to an FEC decoding scheme, and the A parity bits **803** may be combined with one or more portions of the A-field **802** to produce one or more FEC codewords **808** that can be decoded according to the same (or a different) FEC coding scheme. Example suitable FEC codes include Reed-Solomon codes, turbo codes, and LDPC codes, among other examples. With reference to FIG. **3**, each of the FEC codewords **807** and **808** may be one example of the TX codewords **312** (or RX codewords **322**). More specifically, the A parity bits **803** may be used to recover control information carried in the A-field **802** and the B parity bits **805** may be used to recover control information carried in the B-field **804**. In some implementations, the B parity bits **805** may not be used to recover information carried in at least one of the subfields (such as Subfield 0) of the B-field **804**.

(71) As shown in FIG. **8**A, the A parity bits **803** are transmitted (and received) immediately following the A-field **802**, the B-field **804** is transmitted (and received) immediately following the A parity bits **803**, and the B parity bits **805** are transmitted (and received) immediately following the B-field **804**. Compared to the enhanced dummy bearers **500**, **601**, and **700** described with reference to FIGS. **5-7**, the example arrangement of parity bits **803** and **805** allows for earlier decoding of the A-field **802** (such as before the B-field **804**) and correcting of errors therein. However, because the A-field **802** can be ignored by ULE devices operating in the normal mode, the benefits of such early decoding for the A-field **802** may be limited. Further, because the A parity bits **803** are inserted between the A-field **802** and the B-field **804**, the enhanced dummy bearer **800** is not backwards compatible with legacy portable devices.

(72) FIG. **8**B shows another enhanced dummy bearer **810** usable for wireless communications between a base station and a portable device, according to some implementations. In some implementations, the enhanced dummy bearer **810** may be one example of the enhanced dummy bearer **400** of FIG. **4**.

(73) The enhanced dummy bearer **810** includes an S-field **811**, an A-field **812**, a B-field **813**, first X- and Z-fields **814**, one or more “A” parity bits **815**, one or more “B” parity bits **816**, and second X- and Z-fields **817**. With reference to FIG. **4**, the S-field **811**, A-field **812**, and B-field **813** may be examples of the S-field **410**, A-field **420**, and B-field **430**, respectively, of the enhanced dummy bearer **400**. The second X- and Z-fields **817** may be examples of the X-field **440** and the Z-field **450** (shown as a combined block, in FIG. **8**B, for simplicity). Thus, the S-field **811** may carry a packet preamble and synchronization information which marks the start of a DECT packet transmission; the A-field **812** may carry control information associated with a base station or network; the B-field **813** may carry control information associated with a ULE mode of operation by the base station; and the second X- and Z-fields **817** may carry CRC information generated over parts of the B-field **813**.

(74) The A parity bits **815** and the B parity bits **816** may be examples of the parity information **432** of FIG. **4**. In some implementations, the A parity bits **815** may be combined with one or more portions of the A-field **812** to produce one or more FEC codewords **818** that can be decoded according to an FEC decoding scheme, and the B parity bits **816** may be combined with one or

more portions of the B-field **813** to produce one or more FEC codewords **819** that can be decoded according to the same (or a different) FEC coding scheme. With reference to FIG. **3**, each of the FEC codewords **818** and **819** may be one example of the TX codewords **312** (or RX codewords **322**). More specifically, the A parity bits **815** may be used to recover control information carried in the A-field **812** and the B parity bits **816** may be used to recover control information carried in the B-field **813**. In some implementations, the B parity bits **816** may not be used to recover information carried in at least one of the subfields (such as Subfield 0) of the B-field **813**.

(75) In some implementations, the first X- and Z-fields **814** may provide backwards compatibility with legacy devices. With reference for example to FIG. **2**, the first five fields **811-814** of the enhanced dummy bearer **810** match the first five fields **210-250** of the dummy bearer **200**, which conforms with existing DECT ULE standards. In such implementations, the first X- and Z-fields **814** may carry CRC information generated over parts of the B-field **813**. In some other implementations, the first X- and Z-fields **814** may be repurposed to carry other information (such as signaling information for the enhanced dummy bearer **810**). In some implementations, the A parity bits **815** or the B parity bits **816** also may be used to recover information carried in the first X- and Z-fields **814**. In such implementations, the FEC codewords **818** or **819** may include the first X- and Z-fields **814** (in addition to one or more portions of the A-field **812** or the B-field **813**, respectively).

(76) As shown in FIG. **8**B, the A parity bits **815** are transmitted (and received) immediately following the first X- and Z-fields **814**, and the B parity bits **816** are transmitted (and received) immediately following the A parity bits **815**. Compared to the enhanced dummy bearers **500**, **601**, and **700** described with reference to FIGS. **5-7**, the example arrangement of parity bits **815** and **816** allows for earlier decoding of the A-field **812** (such as before the B-field **813**), and correcting of errors therein, while maintaining backwards compatibility with legacy portable devices. However, because the A-field **812** can be ignored by ULE devices operating in the normal mode, the benefits of such early decoding for the A-field **812** may be limited.

(77) FIG. **8**C shows another enhanced dummy bearer **820** usable for wireless communications between a base station and a portable device, according to some implementations. In some implementations, the enhanced dummy bearer **820** may be one example of the enhanced dummy bearer **400** of FIG. **4**.

(78) The enhanced dummy bearer **820** includes an S-field **821**, an A-field **822**, a B-field **823**, first X- and Z-fields **824**, one or more parity bits **825**, and second X- and Z-fields **826**. With reference to FIG. **4**, the S-field **821**, A-field **822**, and B-field **823** may be examples of the S-field **410**, A-field **420**, and B-field **430**, respectively, of the enhanced dummy bearer **400**. The second X- and Z-fields **826** may be examples of the X-field **440** and the Z-field **450** (shown as a combined block, in FIG. **8**C, for simplicity). Thus, the S-field **821** may carry a packet preamble and synchronization information which marks the start of a DECT packet transmission; the A-field **822** may carry control information associated with a base station or network; the B-field **823** may carry control information associated with a ULE mode of operation by the base station; and the second X- and Z-fields **826** may carry CRC information generated over parts of the B-field **823**.

(79) In some implementations, the first X- and Z-fields **824** may provide backwards compatibility with legacy devices. With reference for example to FIG. **2**, the first five fields **821-824** of the enhanced dummy bearer **820** match the first five fields **210-250** of the dummy bearer **200**, which conforms with existing DECT ULE standards. In such implementations, the first X- and Z-fields **824** may carry CRC information generated over parts of the B-field **823**. In some other implementations, the first X- and Z-fields **824** may be repurposed to carry other information (such as signaling information for the enhanced dummy bearer **820**).

(80) The parity bits **825** may be one example of the parity information **432** of FIG. **4**. In some implementations, the parity bits **825** may be combined with the A-field **822**, the B-field **823**, and the first X- and Z-fields **824** to produce one or more FEC codewords **827** that can be decoded

according to an FEC decoding scheme. With reference to FIG. **3**, the FEC codewords **827** may be one example of the TX codewords **312** (or RX codewords **322**). More specifically, the parity bits **825** may be used to recover control information carried in the A-field **822** and the B-field **823** as well as CRC (or signaling) information carried in the first X- and Z-fields **824**.

(81) As shown in FIG. **8**C, the parity bits **825** can be used to decode the A-field **822** and the B-field **823**, collectively, and correct any errors therein. This allows a large portion of the enhanced dummy bearer **820** to be encoded (and decoded) using a single FEC coding operation. Thus, compared to the enhanced dummy bearers **500**, **601**, and **700** described with reference to FIGS. **5-7**, the enhanced dummy bearer **820** may be simpler to implement. However, because the A-field **822** and the B-field **823** are required for FEC decoding, a ULE device cannot begin to receive the enhanced dummy bearer **820** starting at the B-field **823** or terminate its reception of the enhanced dummy bearer **820** prior to receiving all of the parity bits **825** (without sacrificing use of the parity bits **825**).

(82) FIG. **8**D shows another enhanced dummy bearer **830** usable for wireless communications between a base station and a portable device, according to some implementations. In some implementations, the enhanced dummy bearer **830** may be one example of the enhanced dummy bearer **400** of FIG. **4**.

(83) The enhanced dummy bearer **830** includes an S-field **831**, an A-field **832**, a B-field **833**, first X- and Z-fields **824**, one or more “B” parity bits **835**, and second X- and Z-fields **836**. With reference to FIG. **4**, the S-field **831**, A-field **832**, and B-field **833** may be examples of the S-field **410**, A-field **420**, and B-field **430**, respectively, of the enhanced dummy bearer **400**. The second X- and Z-fields **836** may be examples of the X-field **440** and the Z-field **450** (shown as a combined block, in FIG. **8**D, for simplicity). Thus, the S-field **831** may carry a packet preamble and synchronization information which marks the start of a DECT packet transmission; the A-field **832** may carry control information associated with a base station or network; the B-field **833** may carry control information associated with a ULE mode of operation by the base station; and the second X- and Z-fields **836** may carry CRC information generated over parts of the B-field **833**.

(84) The B parity bits **835** may be one example of the parity information **432** of FIG. **4**. In some implementations, the B parity bits **835** may be combined with one or more portions of the B-field **833** to produce one or more FEC codewords **837** that can be decoded according to a FEC coding scheme. With reference to FIG. **3**, the FEC codewords **837** may be one example of the TX codewords **312** (or RX codewords **322**). More specifically, the B parity bits **835** may be used to recover control information carried in the B-field **833**. In some implementations, the B parity bits **835** may not be used to recover information carried in at least one of the subfields (such as Subfield 0) of the B-field **833**.

(85) In some implementations, the first X- and Z-fields **834** may provide backwards compatibility with legacy devices. With reference for example to FIG. **2**, the first five fields **831-834** of the enhanced dummy bearer **830** match the first five fields **210-250** of the dummy bearer **200**, which conforms with existing DECT ULE standards. In such implementations, the first X- and Z-fields **834** may carry CRC information generated over parts of the B-field **833**. In some other implementations, the first X- and Z-fields **834** may be repurposed to carry other information (such as signaling information for the enhanced dummy bearer **830**). In some implementations, the B parity bits **835** also may be used to recover information carried in the first X- and Z-fields **834**. In such implementations, the FEC codewords **837** may include the first X- and Z-fields **834** (in addition to one or more portions of the B-field **833**).

(86) As shown in FIG. **8**D, the enhanced dummy bearer **830** does not support FEC encoding of the A-field **832**. Compared to the enhanced dummy bearers **500**, **601**, and **700** described with reference to FIGS. **5-7**, more parity information can be allocated for FEC encoding the B-field **833** of the enhanced dummy bearer **830**. In other words, the enhanced dummy bearer **830** may include a greater number of B parity bits than any of the enhanced dummy bearers **500**, **601**, or **700**. This

may improve the performance of FEC decoding for the B-field **833** while being simpler to implement. However, because the enhanced dummy bearer **830** does not include any "A" parity bits, a ULE device cannot perform error correction (or FEC decoding) on the A-field **832** during subscription or service call procedures.

(87) Aspects of the present disclosure further recognize that data interleaving (such as be reordering the sequence of bits in a packet) may further improve the FEC coding performance of any of the enhanced dummy bearers **500**, **601**, **700**, and **800-830** described with reference to FIGS. **5-8**D. However, existing versions of the DECT ULE standard do not support data interleaving for dummy bearers. As such, an enhanced dummy bearer with interleaved data may not be backward compatible with legacy devices. Further, because data interleaving changes the order of bits in a packet, a portable device may need to receive an interleaved dummy bearer in its entirety to recover all of the data for any one field. As a result, the portable device may not be able to terminate reception of the dummy bearer early (such as before the end of the packet) or initiate reception of the dummy bearer late (such as after the beginning of the packet).

(88) FIG. **9** shows a block diagram of an example wireless communication device **900**, according to some implementations. In some implementations, the wireless communication device **900** may be one example of a portable device associated with a DECT ULE network (or HAN), such as the portable device **120** of FIG. **1** or the ULE device **608** of FIGS. **6**A and **6**B. With reference for example to FIG. **1**, the portable device may be configured to communicate with a base station over a wireless communication channel (such as a DECT air interface).

(89) The wireless communication device **900** includes a communication interface **910**, a processing system **920**, and a memory **930**. The communication interface **910** is configured to communicate with the base station over the wireless communication channel. In some implementations, the communication interface **910** may receive, over the wireless communication channel, at least a portion of a dummy bearer that includes a B-field followed by one or more parity bits, where the B-field carries control information associated with a ULE mode of operation by a DECT base station.

(90) The memory **930** includes a non-transitory computer-readable medium (including one or more nonvolatile memory elements, such as EPROM, EEPROM, Flash memory, or a hard drive, among other examples) that may store at least the following software (SW) modules: a CW construction SW module **932** to construct a codeword that includes at least part of the B-field and the one or more parity bits; and a FEC decoding SW module **934** to recover the control information based on the codeword and an FEC code associated with the codeword.

Each software module includes instructions that, when executed by the processing system **920**, causes the wireless communication device **900** to perform the corresponding functions.

(91) The processing system **920** may include any suitable one or more processors capable of executing scripts or instructions of one or more software programs stored in the wireless communication device **900** (such as in the memory **930**). For example, the processing system **920** may execute the CW construction SW module **932** to construct a codeword that includes at least part of the B-field and the one or more parity bits. The processing system **920** may further execute the FEC decoding SW module **934** to recover the control information based on the codeword and an FEC code associated with the codeword.

(92) FIG. **10** shows an illustrative flowchart depicting an example operation **1000** for wireless communications, according to some implementations. In some implementations, the example operation **1000** may be performed by a wireless communication device such as the portable device **120** of FIG. **1**, the ULE device **608** of FIGS. **6**A and **6**B, or the wireless communication device **900** of FIG. **9**.

(93) The wireless communication device receives, over a wireless communication channel, at least a portion of a dummy bearer that includes a B-field followed by one or more first parity bits, where the B-field carries control information associated with a ULE mode of operation by a DECT base station (**1010**). The wireless communication device constructs a first codeword that includes at least

part of the B-field and the one or more first parity bits (**1020**). In some implementations, the first codeword may exclude a first subfield of the B-field, where the first subfield signals the start of the B-field. The wireless communication device recovers the control information based on the first codeword and an FEC code associated with the first codeword (**1030**).
(94) In some aspects, the dummy bearer may further include X- and Z-fields following the B-field and preceding the one or more first parity bits. In some implementations, the X- and Z-fields may carry CRC information associated with the B-field. In some other implementations, the X- and Z-fields may carry signaling information associated with the dummy bearer. In some implementations, the first codeword may further include the X- and Z-fields.
(95) In some aspects, the one or more first parity bits may be received immediately after the Z-field. In some other aspects, the dummy bearer may further include one or more non-parity bits between the Z-field and the one or more first parity bits. In some implementations, the one or more non-parity bits may carry signaling information associated with the dummy bearer.
(96) In some aspects, the dummy bearer may further include an A-field carrying legacy control information associated with the DECT base station and one or more second parity bits following the A-field. In some implementations, the one or more second parity may be received after the one or more first parity bits. In some implementations, the wireless communication device may further construct a second codeword that includes at least part of the A-field and the one or more second parity bits and recover the legacy control information based on the second codeword and a FEC code associated with the second codeword.
(97) Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.
(98) Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.
(99) The methods, sequences or algorithms described in connection with the aspects disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor.
(100) In the foregoing specification, embodiments have been described with reference to specific examples thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader scope of the disclosure as set forth in the appended claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A method for wireless communication performed by a wireless communication device, comprising: receiving, over a wireless communication channel, at least a portion of a dummy bearer that includes a B-field followed by one or more first parity bits, the B-field carrying control information associated with an ultra low energy (ULE) mode of operation by a digital enhanced cordless telecommunications (DECT) base station; constructing a first codeword that includes at least part of the B-field and the one or more first parity bits; and recovering the control information based on the first codeword and a forward error correction (FEC) code associated with the first codeword.
2. The method of claim 1, wherein the first codeword excludes a first subfield of the B-field, the first subfield signaling the start of the B-field.
3. The method of claim 1, wherein the dummy bearer further includes X- and Z-fields following the B-field and preceding the one or more first parity bits.
4. The method of claim 3, wherein the X- and Z-fields carry cyclic redundancy check (CRC) information associated with the B-field.
5. The method of claim 3, wherein the X- and Z-fields carry signaling information associated with the dummy bearer.
6. The method of claim 3, wherein the first codeword further includes the X- and Z-fields.
7. The method of claim 3, wherein the one or more first parity bits are received immediately after the Z-field.
8. The method of claim 3, wherein the dummy bearer further includes one or more non-parity bits between the Z-field and the one or more first parity bits.
9. The method of claim 8, wherein the one or more non-parity bits carry signaling information associated with the dummy bearer.
10. The method of claim 1, wherein the dummy bearer further includes an A-field carrying legacy control information associated with the DECT base station and one or more second parity bits following the A-field.
11. The method of claim 10, wherein the one or more second parity are received after the one or more first parity bits.
12. The method of claim 10, further comprising: constructing a second codeword that includes at least part of the A-field and the one or more second parity bits; and recovering the legacy control information based on the second codeword and a FEC code associated with the second codeword.
13. A wireless communication device comprising: a processing system; and a memory storing instructions that, when executed by the processing system, causes the wireless communication device to: receive, over a wireless communication channel, at least a portion of a dummy bearer that includes a B-field followed by one or more first parity bits, the B-field carrying control information associated with an ultra low energy (ULE) mode of operation by a digital enhanced cordless telecommunications (DECT) base station; construct a first codeword that includes at least part of the B-field and the one or more first parity bits; and recover the control information based on the first codeword and a forward error correction (FEC) code associated with the first codeword.
14. The wireless communication device of claim 13, wherein the first codeword excludes a first subfield of the B-field, the first subfield signaling the start of the B-field.
15. The wireless communication device of claim 13, wherein the dummy bearer further includes X- and Z-fields following the B-field and preceding the one or more first parity bits.
16. The wireless communication device of claim 15, wherein the X- and Z-fields carry cyclic redundancy check (CRC) information associated with the B-field.
17. The wireless communication device of claim 15, wherein the X- and Z-fields carry signaling information associated with the dummy bearer.
18. The wireless communication device of claim 15, wherein the first codeword further includes the X- and Z-fields.

19. The wireless communication device of claim 15, wherein the dummy bearer further includes one or more non-parity bits between the Z-field and the one or more first parity bits, the one or more non-parity bits carrying signaling information associated with the dummy bearer.
20. The wireless communication device of claim 13, wherein the dummy bearer further includes an A-field carrying legacy control information associated with the DECT base station and one or more second parity bits following the A-field, execution of the instructions further causing the wireless communication device to: construct a second codeword that includes at least part of the A-field and the one or more second parity bits; and recover the legacy control information based on the second codeword and a FEC code associated with the second codeword.